

Comparative simulated microgravity modalities (3D Clinostat vs. RPM 2.0 vs. 1g Static) in *Mycobacterium marinum* (Clary et al., 2022)

A — Biological Model & Biofilm Substrate

M. marinum 1218R

Flaskette Culture Vessel (31°C, 4 Days)

- ## B — Microgravity Simulation Hardware

Continuous 2-axis clinorotation ($I = 1.5$ rpm, $O = 3.825$ rpm)

2. Random Positioning Machine 2.0 ($n = 3$)

Random velocity vectoring, time-averaged $< 0.01q$

3. Static 1g Earth Control ($n = 3$)

Static incubator shelf adjacent to simulators

C — Integrated FAIR Computational Workflow: WGCNA + Nature 2025 Tabular AI (TabPFN)

OSDR REST API Harvest

5,424 Gene Models

Median-of-Ratios & VST
DESeq2/edgeR Contrasts

Soft Threshold $\beta = 6$

Topological Overlap (TOM)

5 Discrete Modules

Module Eigengenes (MEs)

Prior-Data Transformer

Bayesian In-Context Priors

16 Topological Features

100% LOOCV Modality

Zenodo JSON Schema

RO-Crate v1.1 Spec

Semantic Data Dictionary

Multi-Format Manuscripts

Open Science Ready